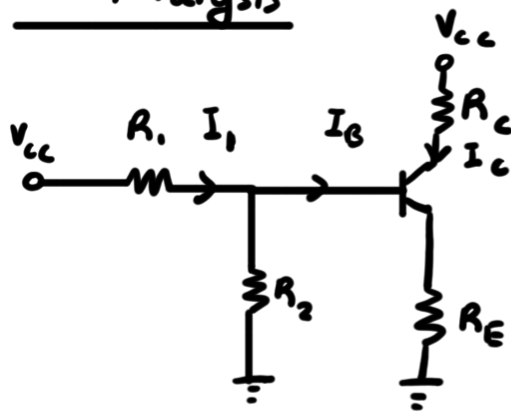


Figure 2(a)

DC Analysis



$$V_{TH} = \frac{R_2}{R_1 + R_2} V_{CC} \quad R_{TH} = R_1 \parallel R_2$$

$$\frac{R_2}{R_1 + R_2} V_{CC} = I_B (R_1 \parallel R_2) + V_{BE} + I_B (1 + \beta) R_E$$

$$I_B = \frac{V_{CC} R_2 / (R_1 + R_2) - V_{BE}}{(R_1 \parallel R_2) + (1 + \beta) R_E}$$

$$I_C = \frac{V_{CC} R_2 / (R_1 + R_2) - V_{BE}}{(R_1 \parallel R_2) + (1 + \beta) R_E} \times \beta$$

$$I_E = \frac{V_{CC} R_2 / (R_1 + R_2) - V_{BE}}{(R_1 \parallel R_2) + (1 + \beta) R_E} \times (\beta + 1)$$

$$V_{CE} = I_C R_C + V_{CE} + I_E R_E$$

$$V_{CE} = V_{CC} - I_C R_C - I_E R_E$$

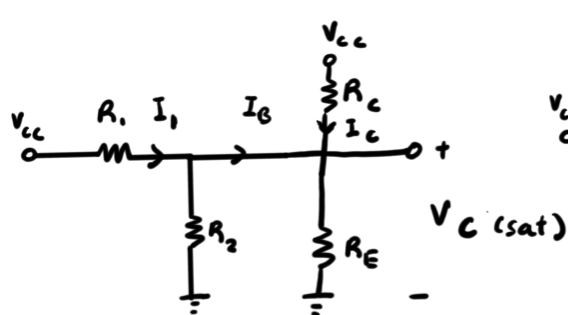
$$= V_{CC} - \frac{V_{CC} R_2 / (R_1 + R_2) - V_{BE}}{(R_1 \parallel R_2) + (1 + \beta) R_E} \times [\beta R_C + (\beta + 1) R_E]$$

$$P_{total} = V_{CC} \times I_{total} \quad (\text{from research})$$

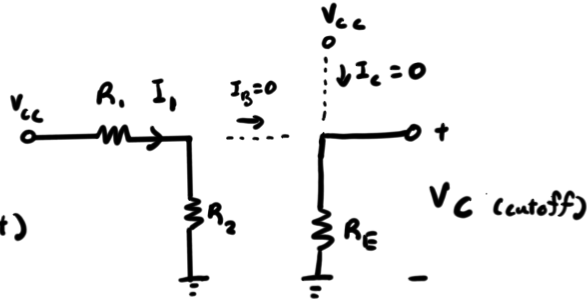
$$= V_{CC} \times (I_1 + I_2)$$

$$= V_{CC} \times \left(\frac{V_{CC}}{R_1 + R_2} + I_C \right) \text{ assume } I_B = 0$$

$$= V_{CC} \left[\frac{V_{CC}}{R_1 + R_2} + \frac{V_{CC} R_2 / (R_1 + R_2) - V_{BE}}{(R_1 \parallel R_2) + (1 + \beta) R_E} \times \beta \right]$$



$$V_{C(sat)} = V_{CC} \frac{R_2 \parallel R_E}{R_2 \parallel R_E + R_1 \parallel R_C}$$

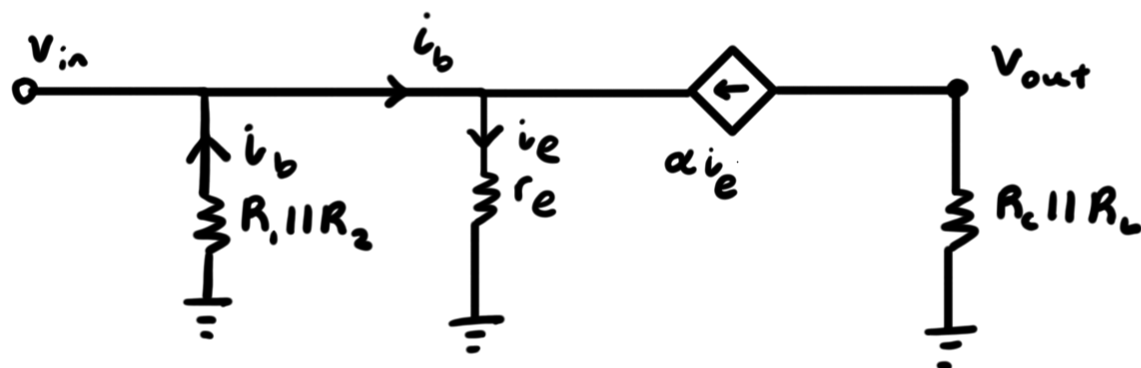


$$V_{C(cutoff)} = V_{CC}$$

$$V_{out} = \min(V_{C(sat)}, V_{C(cutoff)}) - V_{CE} \text{ (from research)}$$

$$\underbrace{V_{swing} = 2V_{out}}_{\text{from research}} = 2 \left\{ V_{CC} \min \left(1, \frac{R_2 \parallel R_E}{R_2 \parallel R_E + R_1 \parallel R_C} \right) - V_{CC} + \frac{V_{CC} R_2 / (R_1 + R_2) - V_{BE}}{(R_1 \parallel R_2) + (1 + \beta) R_E} \times [\beta R_C + (\beta + 1) R_E] \right\}$$

AC Analysis



$$V_{out} = -\alpha i_e (R_C \parallel R_L)$$

$$V_{in} = i_e r_e$$

$$\begin{aligned} A_V = \frac{V_{out}}{V_{in}} &= -\alpha \left(\frac{R_C \parallel R_L}{r_e} \right) \\ &= -\frac{\alpha (R_C \parallel R_L) (I_E)}{V_T} = -\frac{\alpha (R_C \parallel R_L)}{V_T} \times \frac{V_{CC} R_2 / (R_1 + R_2) - V_{BE}}{(R_1 \parallel R_2) + (1 + \beta) R_E} \times (\beta + 1) \end{aligned}$$

$$Z_{out} = R_C \parallel R_L$$

$$Z_{in} = \frac{V_{in}}{i_b} = R_1 \parallel R_2$$

$$BW = f_H - f_L \quad (\text{from research})$$

$$f_L = \max(f_{C_{in}}, f_{C_c}, f_{C_E}) \quad (\text{from research})$$

$$X_{C_{in}} = \frac{1}{2\pi f_{C_{in}} C_{in}} = Z_{in}$$

$$X_{C_c} = \frac{1}{2\pi f_{C_c} C_c} = R_c + R_L$$

$$f_{C_{in}} = \frac{1}{2\pi Z_{in} C_{in}}$$

$$= \frac{1}{2\pi (R_1 || R_2) C_{in}}$$

$$f_{C_c} = \frac{1}{2\pi (R_c + R_L) C_c}$$

$$X_{C_E} = \frac{1}{2\pi f_{C_E} C_E} = r_e$$

$$f_{C_E} = \frac{I_E}{2\pi V_T C_E} = \frac{V_{CC} R_2 / (R_1 + R_2) - V_{BE}}{(R_1 || R_2) + (1 + \beta) R_E} \times \frac{(\beta + 1)}{2\pi V_T C_E}$$

$$f_L = \max\left(\frac{1}{2\pi (R_1 || R_2) C_{in}}, \frac{1}{2\pi (R_c + R_L) C_c}, \frac{V_{CC} R_2 / (R_1 + R_2) - V_{BE}}{(R_1 || R_2) + (1 + \beta) R_E} \times \frac{(\beta + 1)}{2\pi V_T C_E}\right)$$

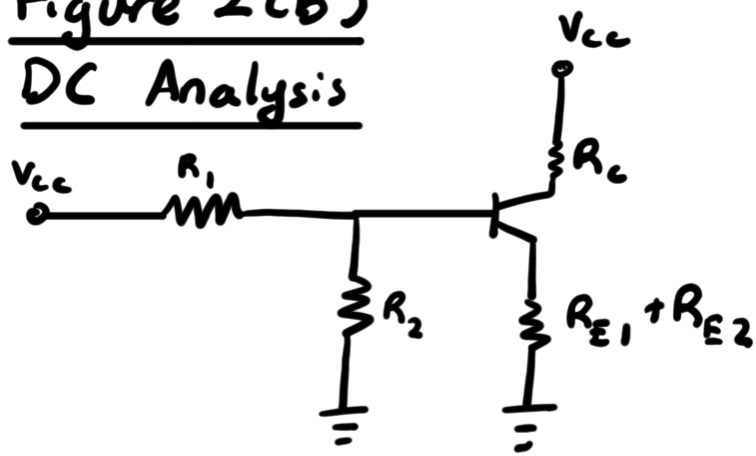
$$f_H = \min(f_{H_i}, f_{H_o}) \quad (\text{from research})$$

$$X_{C_{in}} = \frac{1}{2\pi f_{H_i} C_{in(\text{total})}}$$

WTF???

Figure 2(b)

DC Analysis



$$V_{TH} = \frac{R_2}{R_1 + R_2} V_{CC} \quad R_{TH} = R_1 \parallel R_2$$

$$\frac{R_2}{R_1 + R_2} V_{CC} = I_B (R_1 \parallel R_2) + V_{BE} + I_B (1 + \beta) (R_{E1} + R_{E2})$$

$$I_B = \frac{V_{CC} R_2 / (R_1 + R_2) - V_{BE}}{(R_1 \parallel R_2) + (1 + \beta) (R_{E1} + R_{E2})}$$

$$I_C = \frac{V_{CC} R_2 / (R_1 + R_2) - V_{BE}}{(R_1 \parallel R_2) + (1 + \beta) (R_{E1} + R_{E2})} \times \beta$$

$$I_E = \frac{V_{CC} R_2 / (R_1 + R_2) - V_{BE}}{(R_1 \parallel R_2) + (1 + \beta) (R_{E1} + R_{E2})} \times (\beta + 1)$$

$$V_{CC} = I_C R_C + V_{CE} + I_E (R_{E1} + R_{E2})$$

$$V_{CE} = V_{CC} - I_C R_C - I_E (R_{E1} + R_{E2})$$

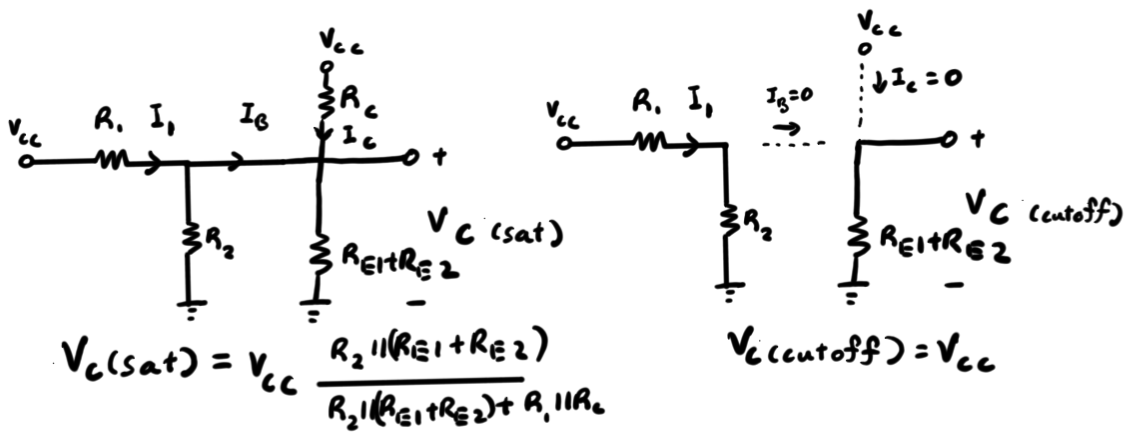
$$= V_{CC} - \frac{V_{CC} R_2 / (R_1 + R_2) - V_{BE}}{(R_1 \parallel R_2) + (1 + \beta) (R_{E1} + R_{E2})} \times [\beta R_C + (\beta + 1) (R_{E1} + R_{E2})]$$

$$P_{total} = V_{CC} \times I_{total} \quad (\text{from research})$$

$$= V_{CC} \times (I_1 + I_2)$$

$$= V_{CC} \times \left(\frac{V_{CC}}{R_1 + R_2} + I_C \right) \text{ assume } I_B = 0$$

$$= V_{CC} \left[\frac{V_{CC}}{R_1 + R_2} + \frac{V_{CC} R_2 / (R_1 + R_2) - V_{BE}}{(R_1 \parallel R_2) + (1 + \beta) (R_{E1} + R_{E2})} \times \beta \right]$$



$$V_{out} = \min(V_{C(sat)}, V_{C(cutoff)}) - V_{CE} \text{ (from research)}$$

$$V_{swing} = 2V_{out} = 2 \left\{ V_{CC} \min \left(1, \frac{R_2 \parallel (R_{E1} + R_{E2})}{R_2 \parallel R_{E1} + R_1 \parallel R_C} \right) - V_{CC} + \frac{V_{CC} R_2 / (R_1 + R_2) - V_{BE}}{(R_1 \parallel R_2) + (1 + \beta) R_E} \times [\beta R_C + (\beta + 1) R_E] \right\}$$

from research

AC Analysis



$$V_{out} = -\alpha i_e (R_C \parallel R_L)$$

$$V_{in} = i_e (r_e + R_{E1})$$

$$A_V = \frac{V_{out}}{V_{in}} = -\alpha \left(\frac{R_C \parallel R_L}{r_e + R_{E1}} \right)$$

$$= -\frac{\alpha (R_C \parallel R_L)}{\frac{V_T}{I_E} + R_{E1}} = -\frac{\alpha (R_C \parallel R_L)}{V_T} \times \frac{V_{CC} R_2 / (R_1 + R_2) - V_{BE}}{(R_1 \parallel R_2) + (1 + \beta) R_E} \times (\beta + 1)$$

to be changed

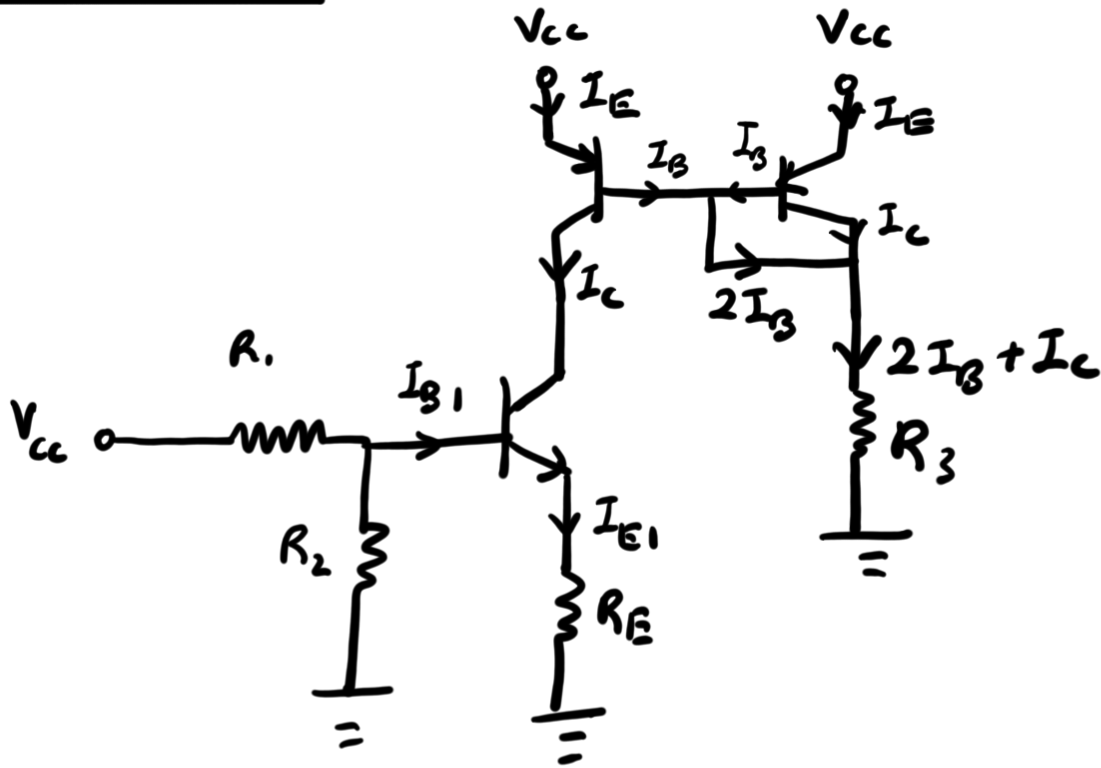
$$Z_{out} = R_C \parallel R_L$$

$$Z_{in} = \frac{V_{in}}{i_b} = R_1 \parallel R_2$$

TO DO: BW

Figure 2(c)

DC Analysis



$$V_{TH} = V_{CC} \left(\frac{R_2}{R_1 + R_2} \right) \quad R_{TH} = R_1 \parallel R_2$$

$$V_{CC} \left(\frac{R_2}{R_1 + R_2} \right) = I_{B1} (R_1 \parallel R_2) + V_{BE} + I_{E1} R_E$$

$$= I_{B1} (R_1 \parallel R_2) + V_{BE} + I_{B1} (1 + \beta) R_E$$

$$I_{B1} = \frac{V_{CC} \left(\frac{R_2}{R_1 + R_2} \right) - V_{BE}}{(R_1 \parallel R_2) + (1 + \beta) R_E}$$

$$I_C = \frac{V_{CC} \left(\frac{R_2}{R_1 + R_2} \right) - V_{BE}}{(R_1 \parallel R_2) + (1 + \beta) R_E} \times \beta = \beta I_{B1} \Rightarrow I_{B1} = I_B$$

$$I_{E1} = \frac{V_{CC} \left(\frac{R_2}{R_1 + R_2} \right) - V_{BE}}{(R_1 \parallel R_2) + (1 + \beta) R_E} \times (\beta + 1)$$

$$I_E = I_B + I_C = I_{B1} (\beta + 1) \Rightarrow I_{E1} = I_E$$

$$P_{total} = V_{CC} \times I_{total}$$

$$= V_{CC} \times (I_1 + I_2 + I_3)$$

$$= V_{CC} \times \left(\frac{V_{CC}}{R_1 + R_2} + 2I_E \right)$$

$$= V_{CC} \times \left(\frac{V_{CC}}{R_1 + R_2} + \frac{V_{CC} \left(\frac{R_2}{R_1 + R_2} \right) - V_{BE}}{(R_1 || R_2) + (1 + \beta) R_E} \times (\beta + 1) \times 2 \right)$$